

# Code Decode Lab - GUI Expansion Pack

## About This Expansion

This GUI version of the Code Decode Lab allows students to interact with encoding and decoding features through a visual interface built with Tkinter.

## What the GUI Includes

- Encoding dropdown menu (utf-8, utf-16, ascii)
- Text input box (accepts strings or byte strings)
- Encode and Decode buttons
- Output display for byte or string results
- Error handling with pop-up messages

## How to Use the GUI

1. Type or paste a string into the input box (e.g. ?????).
2. Select an encoding from the dropdown.
3. Click 'Encode' to convert the string to bytes.
4. To decode, paste a valid Python byte string (e.g. b'\xe3...') into the input box and click 'Decode'.
5. Output will show the result in the lower text area.

## Bonus Challenges

Challenge 1: Add a button to clear both input and output boxes.

Challenge 2: Add support to save the encoded or decoded result to a .txt file.

Challenge 3: Expand the encoding list to include more standards (e.g. cp1252, latin-1).

Challenge 4: Create a 'history' log in the GUI that stores past encodes/decodes.